Information & Management 62 (2025) 104198

Contents lists available at ScienceDirect

Information & Management

journal homepage: www.elsevier.com/locate/im

# Optimal revenue model for Q&A platforms: Paid or free viewership?

Haiyang Feng [a,c], Muzi Li [a], Wenfan Zhao [b,*], Nan Feng [a,c]

[a] College of Management and Economics, Tianjin University, Tianjin 300072, China
[b] Faculty of Humanities and Social Sciences, Beijing Institute of Petrochemical Technology, Beijing 102617, China
[c] Laboratory of Computation and Analytics of Complex Management Systems, Tianjin University, Tianjin 300072, China

ARTICLE INFO

*Keywords:*
Paid Q&A platform
Multi-sided platform
Revenue model
Viewership
Game-theoretic model

ABSTRACT

This paper examines the conditions under which a paid viewership model generates more profits for Q&A platforms compared to a free viewership model. We find that when the unit cost of answer quality is low, the platform should adopt the free viewership model to attract more viewers. Otherwise, the paid viewership model is more profitable, as it incentivizes both askers and answerers to engage more actively. While the answerer consistently benefits more under the paid viewership model, active askers are better off only when the viewership fee remains low.

## 1. Introduction

In recent years, the rapid growth of the digital economy and the widespread use of social networks have significantly increased individuals' demand for acquiring knowledge online. As a result, a wide range of knowledge payment platforms has emerged across sectors such as education, healthcare, and finance. The global paid knowledge market is projected to reach USD 12,954.85 million by 2027.[1] Among the various types of knowledge-based services, online paid Q&A platforms have become particularly prominent.[2] Their usage has surged in areas like online courses and healthcare—especially during the COVID-19 pandemic. According to the 2020 China Internet Medical Industry Research Report, Baidu, together with the Ping An Good Doctor and Chunyu Doctor applications, received over 6.7 million online consultations and >260 million page views during the pandemic. Meanwhile, Dingdang Kuaiyao gained 460,000 users in just one month and served over 6 million people, and Jingdong Health reported an average of 100,000 consultations per day.[3]

On most paid Q&A platforms—such as Zhihu Paid Consultation and Quora Knowledge Prizes—askers pay answerers for personalized responses. Answerers set their own prices, provide answers, and the platform facilitates the exchange, earning a share from each transaction. However, the revenue models differ across platforms. For instance, medical Q&A platforms like Chunyu Doctor and Ping An Good Doctor make all paid answers publicly accessible to all users for free. In contrast, platforms such as Weibo Q&A and Zhihu Paid Consultation charge a small viewership fee (typically 1 RMB) for access to existing answers. This fee is usually split among the platform, the answerer, and the asker, which can increase the total number of content purchases [1].

At different stages of development, platforms may adopt different revenue strategies. For example, Fenda—a leading paid Q&A platform launched in 2016—initially adopted the paid viewership model. However, due to the departure of high-profile answerers, it rebranded as Zaihang Yidian and shifted to a free viewership model. Similarly, Ali Health began with a paid viewership approach but rebranded as Yilu and switched to a free model to attract more users. In contrast, Zhihu Paid Consultation has consistently adhered to the paid viewership model since launching its paid consultation feature and has gradually evolved into a mature system.

Under these two revenue models, the sources of income for platforms and answerers differ. In the free viewership model, the platform's revenue comprises two components: the question revenue generated by active askers and the advertising revenue generated by viewers. The advertising revenue is paid by advertisers who are drawn to the platform's viewer base, as each view represents an opportunity for the platform to display paid advertisements. This reliance on advertising

* Corresponding author.
*E-mail addresses:* hyfeng@tju.edu.cn (H. Feng), limuzicome@tju.edu.cn (M. Li), zhaowenfan@tju.edu.cn (W. Zhao), fengnan@tju.edu.cn (N. Feng).

[1] https://www.linkedin.com/pulse/online-paid-knowledge-market-size-2023-share-jf26f/

[2] A paid Q&A platform is a subset of a knowledge payment platform, with the latter covering a wider range of knowledge-sharing formats such as Q&A content, online courses, tutorials, and so on.

[3] https://www.iimedia.cn/c400/75770.html

https://doi.org/10.1016/j.im.2025.104198
Received 30 January 2024; Received in revised form 8 June 2025; Accepted 10 June 2025
Available online 11 June 2025
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revenue underscores the significant influence of advertisers in shaping the platform's business model, serving as a primary source of income in the absence of direct viewer fees. For the answerer, however, revenue is limited solely to question revenue paid by askers. Conversely, in the paid viewership model, the platform earns from three sources: revenue from active askers' questions, viewership fees from viewers, and advertising revenue. In this model, the platform not only benefits from advertisers but also from direct payments made by viewers through viewership fees. For the answerer, its revenue encompasses both question revenue and viewership fees, reflecting a more diversified income structure. While advertisers still contribute to the platform's overall revenue, the paid model introduces an additional income stream, reducing the platform's dependence on advertising revenue. Hence, the platform encounters varied trade-offs when choosing between different revenue models. Specifically, in the free viewership model, viewers can view existing answers for free, attracting a large number of viewers and generating corresponding advertising revenue. However, there are only two revenue sources for the platform in this model. Conversely, in the paid viewership model, viewers need to pay a fee to view answers, which will discourage some viewers and reduce advertising revenue. However, the platform can also benefit from the viewership fee, creating a third revenue source for the platform. Therefore, this study aims to answer the following research questions: *Which revenue model is more profitable for a Q&A platform? How should an answerer decide the optimal question price and answer quality under different revenue models? How do different revenue models affect the payoffs of the answerer and askers?*

To address these questions, we develop a game-theoretic model in which a Q&A platform connects three-sided users: askers, answerers, and viewers. The platform chooses between two revenue models: the free viewership model, in which viewers can access existing answers at no cost, and the paid viewership model, where viewers must pay to view answers. Under the latter, the viewership revenue is shared among the platform, the answerer, and the asker. The platform's choice of revenue model influences the answerer's decisions regarding question pricing and answer quality. Potential askers can either pay to ask questions for personalized, accurate answers or choose to view existing answers; their decisions depend on comparing the utilities derived from asking versus viewing. Within this framework, we compare the platform's profit, the answerer's payoff, and the surplus gained by active askers under the two revenue models.

We find that the unit cost of answer quality plays a key role in determining which model is more profitable. When this cost is low, the free viewership model yields higher platform profits. When the cost is high, the paid viewership model becomes more profitable. Additionally, under the paid viewership model, the answerer sets a higher question price when the cost of quality is low and provides higher-quality answers when the viewership fee is small. The answerer always earns more under the paid viewership model due to the additional income from viewership fees. However, active askers benefit from the paid model only when the viewership fee is sufficiently low.

Since active askers in the paid viewership model receive a share of the viewership revenue, their number tends to be higher than in the free model. Interestingly, as the unit cost of answer quality increases, both the question price and the quality of answers increase under both revenue models. Moreover, in the paid viewership model, the effects of the viewership fee on the optimal question price and answer quality are non-monotonic: they first increase with the fee, then decrease after a certain point. In the extension model, we introduce a degree of substitutability between different questions and answers on the platform. The results from this extension confirm the robustness of our main findings.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 presents the game-theoretic model and analyzes the decisions faced by all participants. Section 4 provides the equilibrium analysis and compares the platform's profit, the answerer's profit, and the askers' surplus under the two revenue models. Section 5 extends the main model to examine robustness. Section 6 concludes and outlines directions for future research.

## 2. Related literature

Our study investigates how the choice of revenue model—free viewership versus paid viewership—affects the profit of a Q&A platform, the answerer's earnings, and the payoff for askers. Three major streams of literature are relevant to our research.

### 2.1. Knowledge payment platforms and paid Q&A services

The first stream relates to research on online Q&A services and knowledge payment platforms. Several studies have examined the factors influencing users' knowledge contributions in these environments [2–6]. For example, Khansa et al. [3] identify goal-setting artifacts, membership engagement, and habitual behavior as key drivers of participation on Q&A websites. Zhao et al. [5] explore how extrinsic motivation moderates the influence of intrinsic motivation on knowledge sharing in social Q&A communities, finding that extrinsic incentives can undermine users' intrinsic motivation. Shi et al. [6] analyze how prior contributions and community feedback affect subsequent contributor behavior, identifying enactive and vicarious learning as key factors. Other studies focus on improving answer quality [7–8] propose an algorithmic solution to enhance answer quality by adjusting the display order of responses. Peng et al. [9] examine how content-context congruence influences readers' perceptions of value on medical Q&A platforms, highlighting the roles of language style and disease severity as critical cues.

To promote sustainable development, researchers have also investigated new platform functions and operational mechanisms [12–16]. For example, Kuang et al. [13] and Wang et al. [15] assess the impact of monetary incentives on user engagement in knowledge-sharing and online health communities. Shen et al. [16] propose a deep learning–based algorithm that integrates medical knowledge graphs and physician expertise to improve Q&A matching. A growing body of work also addresses platform revenues and market performance [1,17–19]. Ye et al. [1], for instance, identify key factors driving sales on paid Q&A platforms, including social status, endorsements (e.g., likes), and feedback (e.g., comments). Li et al. [17] find that while free-to-paid knowledge differentiation boosts product sales, it may reduce electronic word-of-mouth (eWOM).

While this literature has made significant strides in understanding user behavior, answer quality, and platform functions, relatively few studies address the optimal choice of revenue model for Q&A platforms—especially the comparison between free and paid viewership models. Moreover, limited attention has been paid to how answerers determine optimal pricing and answer quality. Our study addresses these gaps by developing a game-theoretic model to evaluate revenue-sharing mechanisms and identify the conditions under which each revenue model is optimal. This contributes new insights into revenue strategies and pricing decisions for Q&A platforms.

### 2.2. Pricing strategy for information product

The second stream of relevant literature concerns pricing strategies for information products, including bundling [20–24], versioning [25–31], and the freemium models [22,32–36]. Bundling has been widely studied as a way to increase profitability [20]. However, Jin et al. [23] show that piracy can reduce the effectiveness of bundling, highlighting the need for firms to consider piracy risk in bundling decisions. Bhargava [24] addresses the emerging challenge of multi-device content consumption and evaluates the profitability of bundle discounts across various bundling formats (mixed, pure, and partial).

Versioning—offering vertically differentiated products to encourage self-selection—is another critical pricing strategy [28]. Dey and Lahiri [29] incorporate both vertical and horizontal heterogeneity to show

how firms tailor information products to elicit preference diversity. Subsequent work has re-evaluated versioning from an information dissemination perspective, suggesting that consumer underestimation is not necessary for versioning to be effective [30]. The freemium strategy is also prevalent in the information product market [22]. Cheng and Liu [33] compare time-limited and functionality-limited trials to identify when each is more effective. Nan et al. [35] explore freemium strategies in the presence of piracy, finding that under certain consumer perceptions, stronger piracy enforcement may paradoxically harm firms. Other research investigates alternative business models [37–41]. Feng et al. [38] analyze content distribution strategies in platforms offering both B2C and C2C channels. Dou et al. [41] examine whether firms should sell or lease information goods when consumer valuation depreciates over time.

Although prior research offers rich insights into pricing for digital goods, most of it has focused on general markets. Little attention has been paid to Q&A platforms, where the product involves not just content but interactive knowledge services. Our study bridges this gap by incorporating user choices between asking and viewing into the pricing strategy and by analyzing answerers' decisions regarding question pricing and answer quality. This enhances our theoretical understanding of pricing in the context of knowledge platforms and provides practical recommendations for platform design.

### 2.3. Revenue model for two-sided platforms

The third stream of literature examines revenue models for two-sided platforms, which facilitate interactions between distinct user groups [42]. A significant portion of this work addresses pricing structures and revenue allocation [43–47]. Rochet and Tirole [44,45] introduce a foundational model showing that price allocation—not just price levels—determines market outcomes in two-sided platforms. Lin et al. [46] examine the platform's optimal pricing strategy when considering seller price competition and innovation. They find that the degree of variation in buyer-side willingness to pay for quality determines whether or not the platform should subsidize buyers. Alaei et al. [47] compare revenue-sharing rules on music streaming platforms, showing that pro-rata allocation benefits both platforms and artists more than user-centric models.

Another set of studies compares alternative revenue models [27, 48–55]. Casadesus-Masanell and Zhu [49] analyze four business models (i.e., a subscription-based model, an ad-sponsored model, a mixed model, and a dual model) for a high-quality incumbent who faces a low-quality ad-sponsored entrant. Dou et al. [50] evaluate subscription, license-based, and freemium models under strategic buyer behavior. Hagiu and Wright [51] examine when a platform should act as a marketplace or reseller. Chen et al. [52] study how a platform chooses between a brokerage model or an advertising model and how the chosen revenue model affects the payoffs of relevant stakeholders. Zhang et al. [54] assess the performance of data-driven servitization versus traditional product sales.

While a substantial body of work has delved into the pricing structure, revenue allocation, and comparisons of different business strategies across various two-sided platforms, these studies have not adequately accounted for the distinct features of Q&A platforms. The complex interactions among askers, answerers, and viewers, which are fundamental to Q&A platforms, have been overlooked in the existing literature. This study reveals the conditions under which a paid viewership model outperforms a free viewership model in generating profits for Q&A platforms, taking into account the unique interactions among the platform stakeholders. By analyzing these interactions, this study complements existing research on platform revenue models and provides valuable insights into the profitability of Q&A platforms.

## 3. Model setup

We consider a paid Q&A platform that faces three-sided users: askers, answerers, and viewers. The platform facilitates knowledge sharing and diffusion among these participants. In line with the practices of popular platforms such as Zhihu Paid Consultation, Yilu, and Quora Knowledge Prizes, askers are required to pay a fee—set by the answerers—in exchange for accurate responses. We examine two distinct revenue models for the platform: a free viewership model and a paid viewership model. Under the free viewership model, viewers can access answers at no cost, consistent with Yilu's approach. Under the paid viewership model, viewers must pay a flat fee (e.g., 1 RMB) to access existing answers, a model adopted by Zhihu Paid Consultation.

Fig. 1 illustrates the two revenue models employed by paid Q&A platforms. In Fig. 1(a), under the free viewership model, the platform's revenue comprises two components. First, the platform retains a portion of the question fee paid by the asker, with the remainder going to the answerer. Second, the platform earns advertising revenue generated by viewer traffic. In contrast, Fig. 1(b) presents the paid viewership model, where the platform derives revenue from three sources: the question fee, advertising revenue, and a portion of the viewership fee paid by viewers to access existing answers. The remaining viewership fee is equally shared between the answerer and the asker, as depicted in Fig. 2.

### 3.1. Free viewership model

We consider a representative answerer and two distinct user groups on the Q&A platform. The first group comprises potential askers, normalized to a population size of 1. These users decide whether to pose a question or view existing answers based on their perceived utility. The second group consists of potential viewers, also normalized to a size of 1. We assume that each asker submits at most one question on the platform and receives an answer from the answerer at a price $p$, which is set by the answerer. Accordingly, the utility that an asker derives from asking a question is given by

$$u_a = \theta_a q - p + \varepsilon, \tag{1}$$

where $q$ denotes the quality of the answer, $\theta_a$ (with $\theta_a = 1$) represents the degree of matching when a question is asked directly, and $\varepsilon$ captures the emotional value derived from direct interaction with the answer.

On the Q&A platform, potential askers can also choose to view answers closely related to their questions. During this process, they will incur a search cost, denoted by $c_v$. This cost varies across users due to differences in platform familiarity and search behavior, such as the use of different keywords. Therefore, we assume that $c_v$ is uniformly distributed over $[0, 1]$. Thus, a potential asker's utility derived from viewing existing answers is

$$u_v = \theta_v q - c_v, \tag{2}$$

where $\theta_v$ represents the matching degree when viewing existing answers and is defined as $\theta_v = \beta N_a$ ($\beta N_a < 1$), with $\beta$ ($0 < \beta < 1$) denoting the marginal matching degree and $N_a$ indicating the number of active askers, which also corresponds to the number of available answers on the platform. A larger number of answers improves the likelihood that viewers will find relevant content, thereby increasing the matching degree. However, compared to directly asking a question, viewing existing answers generally results in a lower matching degree. In other words, asking a question directly is more likely to yield a precise and accurate response. If $u_a > u_v$, a potential asker will choose to submit a new question; otherwise, viewing existing answers provides a higher utility. Based on Eqs. (1) and (2), the search cost of the marginal asker—who is indifferent between asking a question and viewing existing answers—is given by

$$\widehat{c}_v = p - \varepsilon - q(1 - \beta N_a), \tag{3}$$

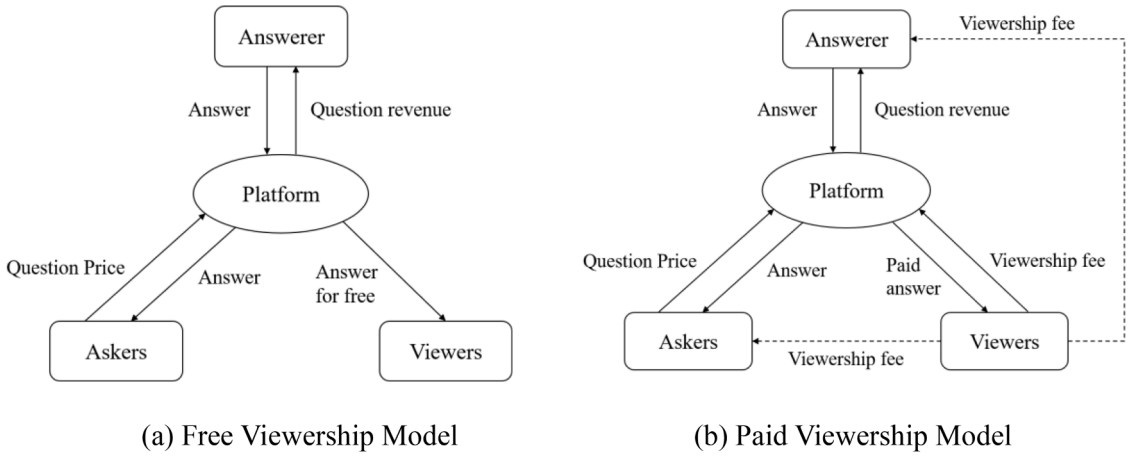



**Fig. 1.** Two revenue models of paid Q&A platforms.

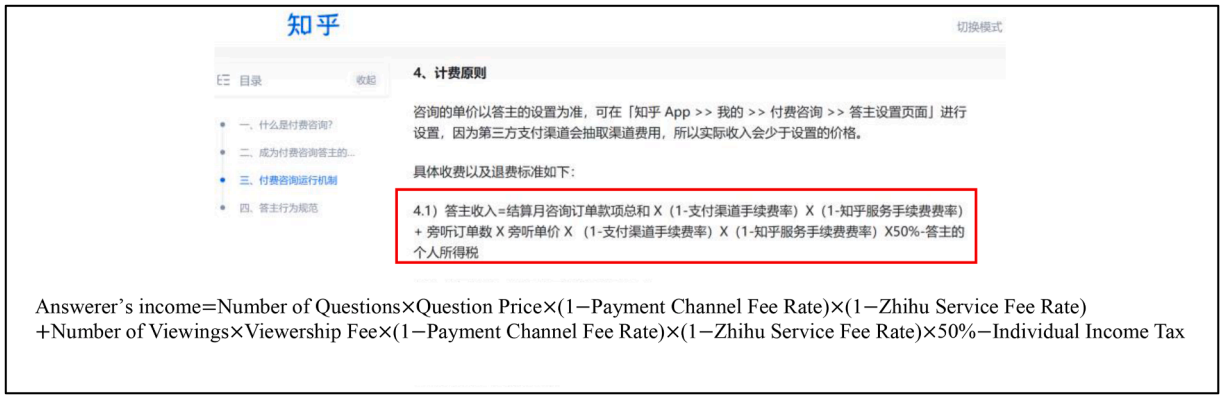



**Fig. 2.** Composition of the answerer's revenue in Zhihu paid consultation.
Note. The payment channel fee rate and the individual income tax are ignored in the present model.

and the number of active askers is

$$N_a = 1 - \widehat{c}_v. \tag{4}$$

Combining Eqs. (3) and (4), we can derive the number of active askers as

$$N_a = \frac{1 - p + q + \varepsilon}{1 + q\beta}. \tag{5}$$

Therefore, the number of potential askers who choose to view existing answers is $1 - N_a$, which we denote as $N_{v1}$ in this study.

The answerer maximizes profit by selecting the optimal question price $p$ and answer quality $q$, given the platform's chosen revenue model. The cost of providing answers may vary due to factors such as time constraints. Let $c$ ($c > 0$) denote the per-unit cost of answer quality. The answerer's profit equals the total revenue from answering questions minus the total cost [56], expressed as

$$\max_{p,q} \pi = N_a(p(1-r) - cq), \tag{6}$$

s.t. $p, q > 0$, $0 \le N_a \le 1$, where $r$ ($0 < r < 1$) is the platform's commission rate, representing the portion of the question price retained by the platform. In addition to revenue from askers, the answer provided may attract viewers from the second user group, thereby generating advertising income for the platform. Let $N_{v2}$ denote the number of such viewers. We assume that the number of viewers per answer is positively related to answer quality, namely,

$$N_{v2} = \alpha q, \tag{7}$$

where $\alpha$ ($\alpha > 0$) represents the number of viewers attracted per unit of answer quality. The platform's total revenue consists of two components: a share of the revenue from answered questions (via commission) and advertising income generated from viewers, as each view represents an opportunity for the platform to earn revenue from paid advertisements. Accordingly, the profit of the platform is

$$\Pi = N_a pr + (N_a N_{v2} + N_{v1})\delta, \tag{8}$$

where $\delta$ denotes the advertising revenue per viewer. In the above equation, the first term represents the commission earned from the answer's question revenue, and the second term represents the advertising revenue.

### 3.2. Paid viewership model

Under the paid viewership model, viewers must pay a viewership fee, $\tau$, to access existing answers. In line with industry practice (as illustrated in Fig. 2), the total revenue generated from viewers is divided among the platform, the asker, and the answerer. Specifically, the platform retains a fraction $r$ of the viewership revenue, while the remaining proportion, $1 - r$, is equally split between the asker and the answerer. Accordingly, the utility an asker derives from posting a question is

$$u_a = \theta_a q - p + \varepsilon + N_{v2}\tau\left(\frac{1-r}{2}\right), \tag{9}$$

where $N_{v2}$ is the number of viewers per answer under the paid viewership model, and $\frac{1-r}{2}$ represents the asker's share of the viewership fee.

We define the number of viewers as

$$N_{v2} = \alpha q - b\tau, \tag{10}$$

where $b$ ($b > 0$) denotes the viewers' price sensitivity, reflecting the degree to which viewership declines as the fee increases. Potential askers also have the option to view existing answers. Considering the viewership fee, the utility derived from viewing is re-formulated as

$$u_v = \theta_v q - c_v - \tau. \tag{11}$$

As in the free viewership model, a potential asker will choose to ask a new question if and only if $u_a > u_v$. Solving this inequality gives the cutoff search cost for the marginal asker: $\widehat{c}_v = \frac{b(1-r)\tau^2 - (2+q(1-r)\alpha)\tau + 2q\beta N_a + 2p - 2(q+\varepsilon)}{2}$. Combining this with Eq. (4), we obtain the number of active askers under the paid viewership model:

$$N_a = \frac{2(1-p+\varepsilon) + \tau(2 - b(1-r)\tau) + q(2 + \alpha\tau - r\alpha\tau)}{2(1+q\beta)}. \tag{12}$$

The answerer earns profit from both the question fee and the share of the viewership revenue. Thus, its profit is

$$\max_{p,q} \pi = N_a(p(1-r) - cq) + N_a N_{v2}\tau\left(\frac{1-r}{2}\right) \tag{13}$$

s.t. $p, q > 0$, $0 \le N_a, N_{v2} \le 1$.

In the above equation, the second term represents the answerer's share of the viewership fee.

Under the paid viewership model, the platform's profit consists of three components: a portion of the question price, advertising revenue from viewers, and a share of the viewership fee. Thus, the platform's profit is given by

$$\Pi = N_a pr + (N_a N_{v2} + N_{v1})(\delta + \tau r). \tag{14}$$

The key notations are summarized in Table 1. The sequence of the game proceeds as follows: First, the platform announces its revenue model (either the free or paid viewership model). Next, the answerer chooses the optimal question price and answer quality. Finally, potential askers decide whether to post questions, and viewers choose whether to view existing answers.

## 4. Equilibrium analysis

In this section, we analyze the equilibrium decisions of potential askers and the answerer, and subsequently derive the equilibrium payoffs for both the answerer and the platform under the free and paid viewership models.

**Table 1**
Summary of notations.

| Variable | Description |
|---|---|
| $p$ | Question price |
| $q$ | Quality of the answer |
| $\theta_a$ ($\theta_v$) | Matching degree when directly asking questions (when viewing answers) |
| $\tau$ | Viewership fee |
| $r$ | Commission rate charged by the platform |
| $N_a$ | Number of users choosing to ask questions among potential askers |
| $N_{v1}$ | Number of users choosing to view answers among potential askers |
| $N_{v2}$ | Number of viewers for each answer |
| $\beta$ | Marginal matching degree |
| $\alpha$ | Number of viewers attracted per unit of answer quality |
| $b$ | Price sensitivity of viewers |
| $c$ | Unit cost of answer quality |
| $c_v$ | Search cost of seeking answers |
| $\delta$ | Unit advertising revenue |
| $\varepsilon$ | Emotional value derived from the direct interaction with answer |

### 4.1. Equilibrium under the free viewership model

When the platform adopts the free viewership model, viewers can access existing answers at no cost. Based on Eqs. (5) and (6), we apply the first-order conditions to derive the equilibrium question price and answer quality. We also derive the equilibrium number of active askers and the answerer's profit under the free viewership model, as summarized in Table 2.

Under the free viewership model, the answerer's income derives solely from questions posed by active askers. The answerer faces a fundamental trade-off when determining the optimal question price and answer quality: increasing the price may discourage users from asking questions, thereby reducing overall earnings despite a higher per-question fee; meanwhile, enhancing answer quality can attract more viewers and active askers but also raises costs, especially since viewers do not contribute revenue under this model. Based on the equilibrium results presented in Table 2, Proposition 1 summarizes how the per-unit cost of quality influences the answerer's strategic decisions.

**Proposition 1.** If the Q&A platform adopts a free viewership model, an increase in the unit cost of answer quality leads to higher optimal question prices and improved answer quality.

*Proof.* All proofs and thresholds are in the appendix.

The cost of providing high-quality answers may vary across answerers due to factors such as time constraints. When the unit cost of quality increases, maintaining the same question price would reduce the answerer's profit. To preserve or enhance profit margins, the answerer is incentivized to raise the question price to offset the higher cost and safeguard their economic interests. According to Eq. (1), however, a higher price discourages some users from asking questions. To counteract this effect and attract more askers, the answerer improves the quality of their answers. Consider the example of a doctor. A highly qualified physician often has demanding hospital duties, resulting in a high opportunity cost for participating in Q&A platforms. In such cases, if the doctor chooses to engage with the platform, the answers provided are priced higher to reflect the time cost, and the quality of the responses is likewise elevated.

### 4.2. Equilibrium under the paid viewership model

When the platform adopts the paid viewership model, viewers are required to pay a fee to access existing answers. Similar to the free viewership model, we apply backward induction to derive the answerer's equilibrium decisions and corresponding profit under the paid viewership model, as summarized in Table 3.

After accounting for the viewership fee, the trade-off faced by the answerer under the paid viewership model differs from that under the free viewership model. Under the free viewership model, only active askers generate revenue for the answerer. In contrast, under the paid viewership model, both active askers and viewers contribute to the answerer's income. Additionally, active askers also benefit from the viewership fee, making it a critical factor in shaping the equilibrium decisions of all participants. The following observation illustrates how the viewership fee influences the answerer's optimal question price and answer quality.

**Table 2**
Equilibrium solutions under the free viewership model.

| Variable | The Answerer's Equilibrium Decisions and Profit |
|---|---|
| Question price | $p_F^* = \frac{c^2 + (1-r)^2(\beta + \beta\varepsilon - 1)}{(1-r)(1-c-r)\beta}$ |
| Answer quality | $q_F^* = \frac{2c + (1-r)(\beta + \beta\varepsilon - 2)}{(1-c-r)\beta}$ |
| Number of active askers | $N_{aF}^* = \frac{1-c-r}{(1-r)\beta}$ |
| Profit | $\pi_F^* = \frac{(1-c-r)(c + (1-r)(\beta + \beta\varepsilon - 1))}{(1-r)\beta^2}$ |

**Table 3**
Equilibrium solutions under the paid viewership model.

| Variable | The Answerer's Equilibrium Decisions and Profit |
|---|---|
| Question price | $p_P^* = \frac{M}{2(1-r)\beta(2c-(1-r)(2+(2-r)\alpha\tau))}$ |
| Answer quality | $q_P^* = \frac{4c-(1-r)(4+2(2-r)\alpha\tau-\beta(2+2\varepsilon+\tau(2-b(2-r)\tau)))}{2(1-c-r)\beta+(2-r)(1-r)\alpha\beta\tau}$ |
| Number of active askers | $N_{aP}^* = \frac{2(1-r)-2c+(1-r)(2-r)\alpha\tau}{2\beta(1-r)}$ |
| Profit | $\pi_P^* = \frac{(2c-(1-r)(2+(2-r)\alpha\tau))(-2c+(1-r)(2+(2-r)\alpha\tau-\beta(2+2\varepsilon+\tau(2-b(2-r)\tau))))}{4(1-r)\beta^2}$ |

Note: $M = 2c((1-r)\tau(2\alpha+b\beta\tau)-2c)+(1-r)^2\big((2+(2-r)\alpha\tau)(2-r\alpha\tau)-\beta\big(2\big(2(1+\varepsilon)+2\tau+(1-r)\big(\tau\alpha(1+\varepsilon)+(\alpha-b)\tau^2\big)\big)+b(2-r)r\alpha\tau^3\big)\big)$.

**Observation 1.** *Under the paid viewership model, as the viewership fee increases, both the question price and the answer quality first rise and then fall.*

Fig. 3

Observation 1 reveals a noteworthy non-monotonic relationship between the viewership fee and the answerer's optimal decisions. Initially, as the viewership fee increases, the answerer recognizes a stronger incentive to provide high-quality answers in order to attract more viewers and maximize earnings. In response, the answerer allocates greater effort to improving answer quality. Active askers are also willing to pay a higher price for these enhanced answers, knowing they not only improve their personal utility but also generate shared viewership revenue. This incentivizes the answerer to raise the question price.

However, as the viewership fee continues to rise, the number of viewers willing to pay diminishes, reducing the answerer's potential revenue. To limit losses, the answerer may lower the quality of its answers, thereby reducing associated costs. In turn, the answerer must also decrease the question price to remain attractive to askers in a shrinking market.

Similar to the free viewership model, an increase in the unit cost of quality under the paid viewership model leads to higher equilibrium levels of both question price and answer quality. When the cost of delivering quality rises, the answerer must invest more resources to maintain answer quality, which directly affects viewer attraction. While a higher viewership fee may reduce individual viewing demand, improved answer quality can still boost total viewership, thereby increasing overall viewership revenue. Simultaneously, increasing the question price helps offset the rising cost of quality provision. The combined effect of these factors leads to upward adjustments in both price and quality. Therefore, Proposition 1 remains valid under the paid viewership model.

### 4.3. Equilibrium comparison

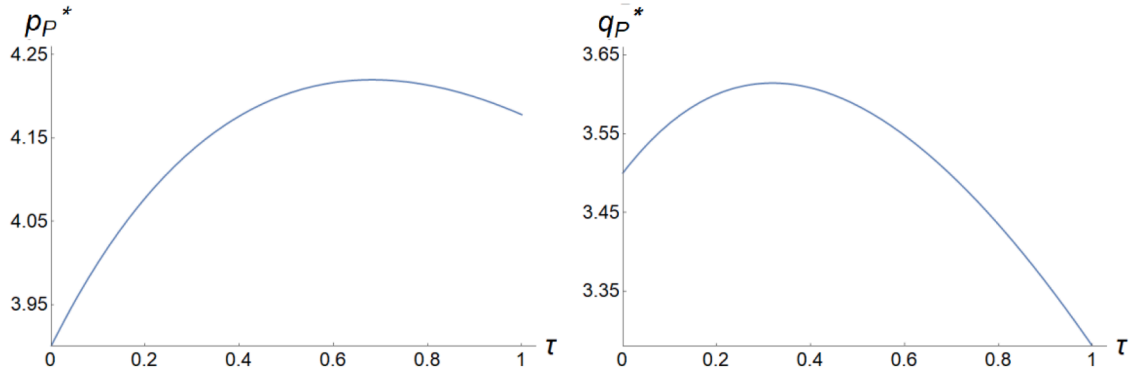
In this section, we first compare the answerer's equilibrium decisions under the two revenue models. We then analyze the platform's profits and identify the conditions under which the paid viewership model generates higher profits than the free viewership model, and vice versa. Finally, we compare the profits and surpluses obtained by the answerer and active askers under both models.

A direct comparison of the optimal question prices in Tables 2 and 3 leads to the following proposition.

**Proposition 2.** If the unit cost of answer quality is low ($c < c_1$), the answerer sets a higher question price under the paid viewership model than under the free viewership model. Conversely, if the unit cost is high ($c \geq c_1$), the optimal question price is higher under the free viewership model.

When the unit cost of answer quality is low, the answerer can provide high-quality answers at relatively low expense. Under the paid viewership model, the existence of a viewership fee incentivizes the answerer to improve answer quality in order to attract more viewers and earn a greater share of the viewership revenue. In this case, the answerer tends to set a relatively high question price. In contrast, when the unit cost of answer quality is high, producing high-quality answers becomes significantly more costly. Under the free viewership model—where viewers pay nothing—the answerer cannot be compensated through viewership fees. To offset the high unit cost and maintain profitability, the answerer must raise the question price substantially. This result is illustrated in Fig. 4. In addition to setting the question price, the answerer must also determine the quality level of the answer. The following proposition summarizes the equilibrium comparison of answer quality.

**Proposition 3.** The answer quality is higher under the paid viewership model if and only if the viewership fee is low ($\tau < \tau_1$).




(a) Impact of $\tau$ on $p_P^*$ (b) Impact of $\tau$ on $q_P^*$

**Fig. 3.** Impact of viewership fee on question price and answer quality.
*Note.* $r = 0.5$, $b = 0.3$, $c = 0.4$, $\alpha = 0.15$, $\beta = 0.5$, $\delta = 1$, $\varepsilon = 0.5$.

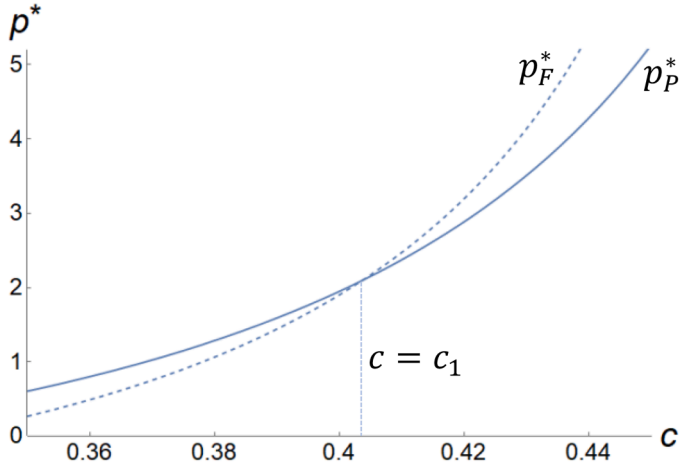



**Fig. 4.** Question price comparison under the two revenue models.
*Note.* $r = 0.5,\ b = 0.25,\ \alpha = 0.25,\ \beta = 0.5, \tau = 0.3,\ \delta = 1,\ \varepsilon = 0.1$.

At first glance, one might expect that the presence of a viewership fee in the paid viewership model would consistently incentivize the answerer to provide higher-quality answers. However, as illustrated in Fig. 5, answer quality is higher under the paid viewership model only when the viewership fee is relatively low. When the viewership fee is low, the cost for users to access existing answers remains minimal, making it easier to attract a large number of viewers. In this case, the answerer is motivated to provide higher-quality answers to draw more viewers and thereby earn a larger share of the viewership revenue. In contrast, under the free viewership model—where viewers pay nothing—the answerer cannot earn additional income from viewers and may therefore be more inclined to limit answer quality to reduce costs.

However, when the viewership fee becomes high, users' willingness to view answers declines in the paid viewership model, reducing the overall number of viewers. Although a higher viewership fee increases income per viewer, the sharp drop in total viewers may lower overall revenue, weakening the answerer's incentive to maintain high answer quality. In contrast, under the free viewership model, viewer behavior and the answerer's income are not influenced by viewership fees. As a result, the answerer may focus more on attracting active askers by enhancing answer quality to maximize earnings from question payments.

Based on the equilibrium decisions of the answerer shown in Tables 2 and 3, we can derive the platform's optimal profits under the two revenue models (see Appendices A.1 and A.2). By comparing these maximum profits, Proposition 4 identifies the platform's optimal choice of revenue model.

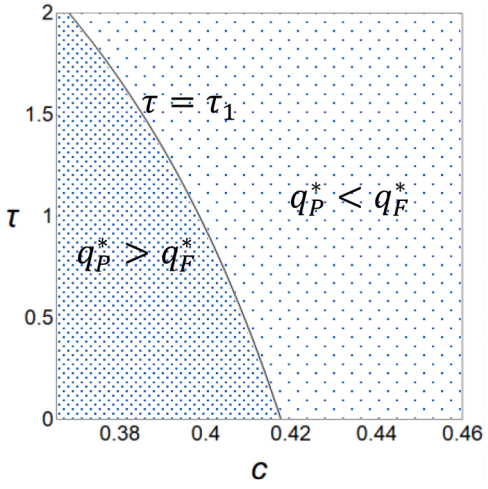



**Fig. 5.** Answer quality comparison under the two revenue models.
*Note.* $r = 0.5,\ b = 0.25,\ \alpha = 0.2,\ \beta = 0.5, \delta = 1,\ \varepsilon = 0.1$.

**Proposition 4.** *When the unit cost of answer quality is low ($c < c_2$), the free viewership model yields higher profit for the platform than the paid viewership model. Conversely, when the unit cost of answer quality is high ($c \geq c_2$), the paid viewership model becomes more profitable for the platform.*

Proposition 4 suggests that when the unit cost of answer quality is low, the platform should adopt the free viewership model. As noted in Proposition 2, a low unit cost of quality allows the answerer to charge a higher question price under the paid viewership model, which can reduce the willingness of active askers to pay. Additionally, the presence of a viewership fee in the paid model may discourage users from viewing answers, thereby reducing the platform's earnings from viewership-related revenue streams. In contrast, under the free viewership model, users can access answers without cost. The low unit cost enables the answerer to provide high-quality responses, attracting more viewers and increasing the platform's advertising revenue. Although the answerer charges a relatively lower question price in this model, the resulting growth in advertising revenue more than compensates for the reduced direct income, leading to higher overall platform profit.

When the unit cost of quality is high, the answerer must charge a significantly higher question price in the free viewership model, which lowers the number of active askers. This, in turn, reduces the number of answers generated and diminishes viewer traffic, negatively affecting advertising revenue. Under the paid viewership model, while the viewership fee may deter some users, the answerer typically sets a lower question price, maintaining a reasonable level of participation among askers. Moreover, the platform earns revenue from multiple sources—including its share of the viewership fees. As the answerer responds to higher costs by improving answer quality (see Proposition 1), the platform benefits from both enhanced viewer willingness to pay and greater revenue from shared fees. The resulting gain often exceeds the loss in user volume under the free model, making the paid viewership model more profitable when quality costs are high. Fig. 6 illustrates the comparison result.

This finding aligns with industry practices. For instance, Baidu Knows relies on volunteer contributors who provide knowledge at relatively low cost, enabling the platform to offer free access to content. Conversely, Xueqiu, which focuses on financial investment, hosts professional answerers who invest significant time and expertise in producing in-depth market analyses and stock research. To access this high-value content, users are required to pay a viewership fee.

Given the platform's choice of revenue model, the answerer determines the optimal question price and answer quality, while potential askers decide whether to pose questions. Based on the answerer's profits and equilibrium decisions derived in Sections 4.1 and 4.2, Proposition 5 summarizes how the revenue model impacts the number of active

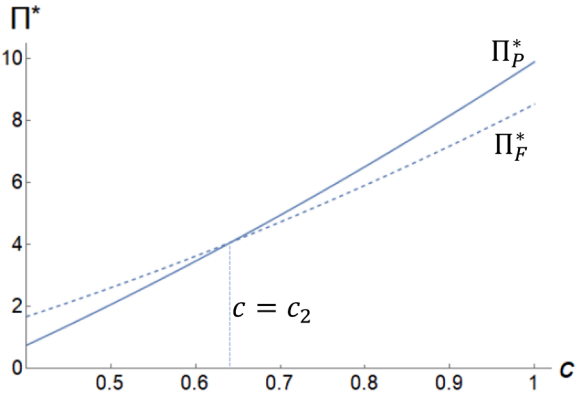



**Fig. 6.** Profit comparison of the platform under the two revenue models.
*Note.* $r = 0.45,\ b = 0.5,\ \alpha = 0.4,\ \beta = 0.6, \tau = 1, \delta = 0.8,\ \varepsilon = 0.8$.

askers, the answerer's profit, and the surplus of these active askers.

Proposition 5.

(a) The number of active askers is higher under the paid viewership model than under the free viewership model.
(b) The answerer earns greater profit under the paid viewership model.
(c) If the viewership fee is low ($\tau < \tau_2$), active askers receive a higher surplus under the paid viewership model; otherwise ($\tau \geq \tau_2$), they are better off under the free viewership model.

In the free viewership model, active askers must pay to receive a perfectly matched answer, while viewers can access content without any fee. Although the answers viewers receive may not be perfectly tailored, users with lower quality expectations or a limited willingness to pay may still prefer to passively view existing answers. As a result, the number of active askers remains relatively limited. In contrast, under the paid viewership model, active askers not only receive accurate answers but can also earn a share of the viewership fees paid by others. This additional incentive increases users' willingness to ask questions directly.

According to Proposition 5 (b) and 5 (c), the platform's choice of revenue model has an asymmetric impact on the payoffs of answerers and active askers. The paid viewership model consistently yields higher profits for the answerer, as it allows them to earn income both from question fees and viewership fees. However, for active askers, the benefits are conditional: they enjoy a higher surplus under the paid viewership model only when the viewership fee is low. The viewership fee directly influences the number of viewers in the paid model. When the fee is relatively low, more users are willing to pay to view answers. This means active askers benefit not only from receiving high-quality responses and emotional satisfaction but also from shared revenue generated by viewers. In contrast, under the free viewership model, they only receive the answer and emotional value—without any monetary return. Therefore, when the viewership fee is low, active askers achieve a higher surplus in the paid viewership model, as illustrated in Fig. 7.

## 5. Considering substitutability between answers

In the main model, we assume that the questions posted by active askers are highly differentiated, with no substitutability between answers. In this section, we extend the analysis by considering the inflation of advertising revenue brought by viewers and incorporating a degree of substitutability among different questions and answers.

Unlike the main model—where the total number of viewers attracted by existing answers is given by $N_a N_{v2}$—we now assume that the overall number of viewers is represented by $N_a(1-\lambda N_a)N_{v2}$, which decreases with $N_a$ when the number of answers on the platform becomes sufficiently large. This adjustment reflects the idea that, due to overlapping content and substitutability, the marginal attractiveness of additional answers diminishes as the platform becomes saturated. Under this setting, the profit function of the answerer remains unchanged. However, the platform's profit under the free viewership model is modified as

$$\widehat{\Pi}_F = N_a p r + (N_a(1-\lambda N_a)N_{v2} + N_{v1})\delta. \tag{15}$$

Similarly, the answerer's profit and the platform's profit under the paid viewership model are now given by

$$\max_{p,q} \widehat{\pi}_P = N_a(p(1-r) - cq) + N_a(1-\lambda N_a)N_{v2}\tau\left(\frac{1-r}{2}\right), \tag{16}$$

and

$$\widehat{\Pi}_P = N_a p r + (N_a(1-\lambda N_a)N_{v2} + N_{v1})(\delta + \tau r). \tag{17}$$

respectively.

Similarly, using backward induction, we derive the equilibrium decisions and payoffs of active askers, the answerer, and the platform under both revenue models when a degree of substitutability exists between different questions and answers (as shown in Appendix A.10). By comparing the answerer's optimal question price and answer quality under the free and paid viewership models, we verify that the conclusions presented in Propositions 2 and 3 of the main model remain robust. This is illustrated in Figs. 8 and 9.

Next, we examine the equilibrium profits of both the platform and the answerer. A comparison of Eqs. (6) and (16) shows that the answerer earns higher profits under the paid viewership model than under the free viewership model, consistent with Proposition 5 (b) in the main model. Similarly, comparing Eqs. (15) and (17), we observe that the platform prefers the paid viewership model only when the unit cost of answer quality is sufficiently high, as illustrated in Fig. 10. This finding confirms the robustness of Proposition 4 in the main model.

Finally, we examine the number of active askers on the platform and their total surplus. Based on the equilibrium number of active askers $\widehat{N}^*_{aF}$ and $\widehat{N}^*_{aP}$ (as detailed in Appendix A.10), more users are willing to directly ask questions under the paid viewership model, consistent with the result in Proposition 5 (a). Furthermore, Fig. 11 compares the surplus of active askers under the two revenue models. It shows that only when the viewership fee is low do active askers benefit more under the paid viewership model. This finding confirms the robustness of Proposition 5 (c).

## 6. Discussion and conclusions

In this paper, we investigate how the choice of revenue

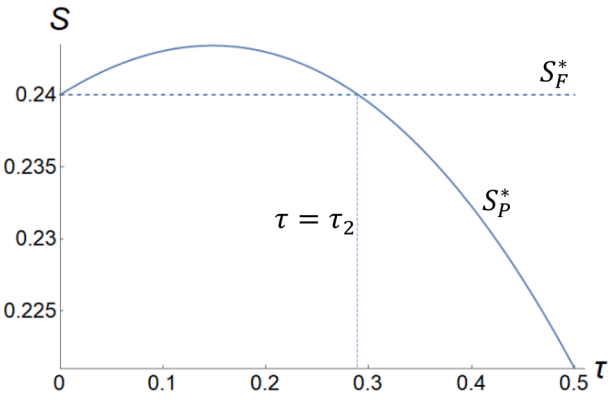



**Fig. 7.** Comparison of active asker surplus under the two revenue models.
*Note.* $r = 0.5,\ b = 0.3,\ c = 0.4, \alpha = 0.15,\ \beta = 0.5, \delta = 1,\ \varepsilon = 1.$

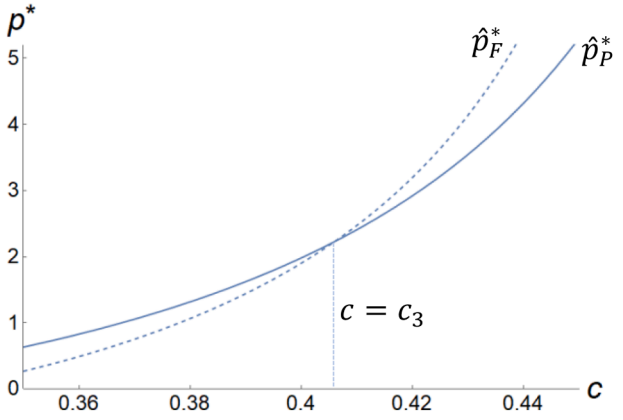



**Fig. 8.** Question price comparison when considering answer substitutability.
*Note.* $r = 0.5,\ b = 0.25,\ \alpha = 0.25,\ \beta = 0.5, \tau = 0.3,\ \delta = 1,\ \varepsilon = 0.1,\ \lambda = 0.1.$

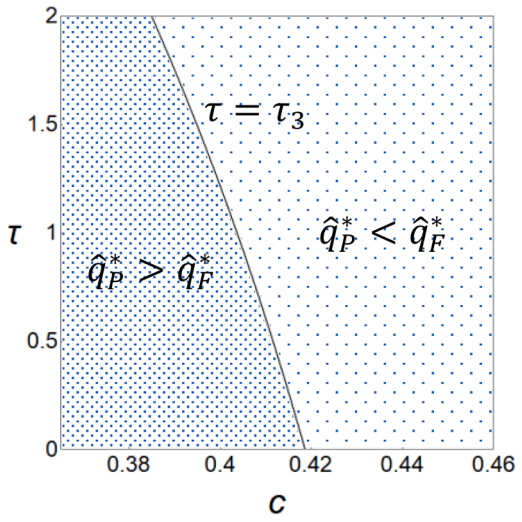



**Fig. 9.** Answer quality comparison when considering answer substitutability.
*Note.* $r = 0.5,\ b = 0.25,\ \alpha = 0.2,\ \beta = 0.5, \delta = 1,\ \varepsilon = 0.1,\ \lambda = 0.1$.

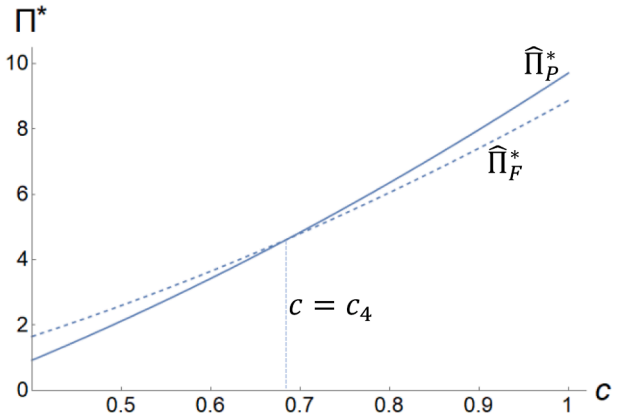



**Fig. 10.** Profit comparison of the platform when considering answer substitutability.
*Note.* $r = 0.45,\ b = 0.5,\ \alpha = 0.4,\ \beta = 0.6, \tau = 1, \delta = 0.8,\ \varepsilon = 0.8,\ \lambda = 0.1$.

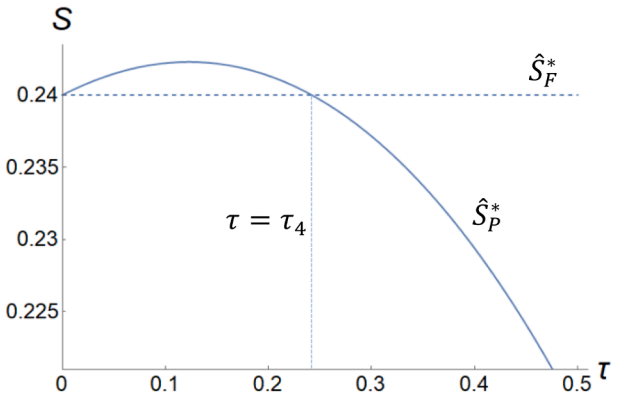



**Fig. 11.** Comparison of active asker surplus when considering answer substitutability.
*Note.* $r = 0.5,\ b = 0.3,\ c = 0.4, \alpha = 0.15,\ \beta = 0.5, \delta = 1,\ \varepsilon = 1,\ \lambda = 0.1$.

model—either the free viewership model or the paid viewership model—affects the profitability of a paid Q&A platform and the payoffs of both answerers and askers. We analyze the answerer's optimal decisions regarding question pricing and answer quality, as well as the askers' decisions on whether to submit questions under different revenue models. We also conduct an extension to verify the robustness of our findings.

We develop a game-theoretic model to evaluate the incentives faced by the three-sided Q&A platform in deciding between the two revenue models. Our results show that when the unit cost of answer quality is low, the platform should adopt the free viewership model; otherwise, the paid viewership model yields higher profits. The answerer consistently earns more profit under the paid viewership model. However, active askers benefit more from this model only if the viewership fee is low. Regarding the answerer's optimal decisions, a key insight is that the paid viewership model does not always incentivize higher question prices or better answer quality. Specifically, the question price is higher under the paid viewership model only when the unit cost of answer quality is low, and answer quality is higher only when the viewership fee is low. Interestingly, as the unit cost of answer quality increases, both the optimal question price and answer quality rise under both revenue models. Furthermore, under the paid viewership model, the relationship between the viewership fee and both question price and answer quality is non-monotonic—initially increasing, then decreasing.

These findings yield several managerial implications. First, for Q&A platforms, it is essential to continuously monitor and assess the cost of providing high-quality answers. This includes factors such as the answerers' time, expertise, and the resources they invest in providing reliable answers. The platform should be flexible in switching between free and paid viewership models based on changes in answerer costs. When costs are low, a free viewership model can attract more askers and viewers. When costs rise, a paid model may be more suitable for maintaining platform profitability. A real-world example of this dynamic is Quora's evolution. Initially, Quora adopted a free viewership model to grow its user base. Later, as the demand for high-quality content increased, Quora introduced Quora+, a subscription-based paid viewership model, to support premium content and incentivize top contributors. Second, for answerers, it is important to adjust question prices and answer quality based on the platform's revenue model, viewership fee level, and their own cost of providing answers. A high cost of answer quality does not necessarily lead to lower quality; the cost can be offset by a higher question price, which in turn motivates higher-quality content. All else being equal, answerers may benefit more from joining platforms with paid viewership models. Third, for active askers, although the paid viewership model offers the potential to earn a share of the viewership fee, they may not always be better off in this model. When the viewership fee is high, the free viewership model may provide them with greater surplus. Therefore, askers should consider the platform's fee structure when deciding whether to join a free- or paid-viewership platform.

Our study suggests several directions for future research. First, this study assumes that the platform adopts a uniform revenue model across all content. In practice, some platforms use a hybrid approach, where certain answers are available for free while others are paywalled. The decision regarding which answers to monetize may lie with the platform, the answerer, or a combination of stakeholders. Although such a hybrid model offers a more flexible monetization strategy, it also introduces substantial complexity in modeling and decision-making, which we leave for future exploration. Second, our model does not account for the entry decision of answerers. As a future direction, we could examine how the entry behavior of answerers influences the platform's choice of revenue model. Finally, the role of advertisers—who contribute to the platform's revenue, particularly under the free viewership model—and their strategic decisions represent another valuable direction for future study.

## CRediT authorship contribution statement

**Haiyang Feng:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Funding

acquisition, Conceptualization. **Muzi Li:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Wenfan Zhao:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis. **Nan Feng:** Supervision, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Acknowledgements

This work was supported by the National Natural Science Foundation of China (NO 72394373 and NO 72231004).

## Appendix

### A.1 Proof of Equilibrium solutions under the free viewership model

*Proof.* When a Q&A platform adopts the free viewership model, based on Eqs. (1) and (2), the location of the user who is indifferent between asking questions and viewing answers is $\widehat{c}_v = p - q - \varepsilon + q\beta N_a$. Thus, the number of active askers is $N_a = \frac{1-p+q+\varepsilon}{1+q\beta}$ (as derived in Eq. (5)). Substitute $N_a$ into Eq. (6) and derive the first-order derivative of $\pi$ with respect to $p$ and $q$. Solving $\frac{\partial \pi}{\partial p} = 0$ and $\frac{\partial \pi}{\partial q} = 0$ yields $p_F^* = \frac{c^2+(1-r)^2(\beta+\beta\varepsilon-1)}{(1-r)(1-c-r)\beta}$ and $q_F^* = \frac{2c+(1-r)(\beta+\beta\varepsilon-2)}{(1-c-r)\beta}$. Substituting $p_F^*$ and $q_F^*$ into Eqs. (5)-(6) yields $N_{aF}^* = \frac{1-c-r}{(1-r)\beta}$, $\pi_F^* = \frac{(1-c-r)(c+(1-r)(\beta+\beta\varepsilon-1))}{(1-r)\beta^2}$, and $\Pi_F^* = \big(r^3(\beta+\beta\varepsilon-1) + r^2(2-2\alpha\delta-2\beta(1+\varepsilon)+\beta\delta(-1+\alpha+\beta+\alpha\varepsilon)) + \delta(\beta(-1+c+\beta)+\alpha(-2+2c+\beta+\beta\varepsilon)) + r\big(-1+c^2+4\alpha\delta-c(2\alpha+\beta)\delta+\beta(1+\varepsilon-2\delta(-1+\alpha+\beta+\alpha\varepsilon))\big)\big)/\big((1-r)^2\beta^2\big)$.

### A.2 Proof of Equilibrium solutions under the paid viewership model

*Proof.* When the Q&A platform adopts the paid viewership model, based on Eqs. (9)- (11), the location of the indifferent user is $\widehat{c}_v = \frac{(2(p-q-\varepsilon)-(2+q(1-r)\alpha)\tau+b(1-r)\tau^2+2q\beta N_a)}{2}$. According to Eq. (4), the number of active askers is $N_a = \frac{2(1-p+\varepsilon)+\tau(2-b\tau(1-r))+q(2+\alpha\tau(1-r))}{2+2q\beta}$. Similar to the Proof of Lemma 1, based on the first-order derivative of $\pi$ (i.e., Eq. (13)) with respect to $p$ and $q$, we have $p_P^* = \big(2c(1-r)\tau(2\alpha+b\beta\tau)-4c^2+(1-r)^2\big((2+(2-r)\alpha\tau)(2-r\alpha\tau)-\beta\big(4(1+\varepsilon)+2(2+(1-r)\alpha(1+\varepsilon))\tau+2(1-r)(\alpha-b)\tau^2+b(2-r)r\alpha\tau^3\big)\big)\big)/(2(1-r)\beta(2c-(1-r)(2+(2-r)\alpha\tau)))$ and $q_P^* = (4c-(1-r)(4+2(2-r)\alpha\tau-\beta(2+2\varepsilon+\tau(2-b(2-r)\tau))))/(2(1-c-r)\beta+(2-r)(1-r)\alpha\beta\tau)$. Substituting $p_P^*$ and $q_P^*$ into Eqs. (12)-(13) yields $N_{aP}^* = \frac{2(1-r)-2c+(1-r)(2-r)\alpha\tau}{2\beta(1-r)}$, $\pi_P^* = \frac{(2c-(1-r)(2+(2-r)\alpha\tau))(-2c+(1-r)(2+(2-r)\alpha\tau-\beta(2+2\varepsilon+\tau(2-b(2-r)\tau))))}{4(1-r)\beta^2}$, and $\Pi_P^* = \frac{1}{4(1-r)^2\beta^2}\big(4\big(r^3(\beta+\beta\varepsilon-1)+r^2(2-2\alpha\delta-2\beta(1+\varepsilon)-\beta\delta(1-\alpha-\beta-\alpha\varepsilon))+\delta(\alpha(2c+\beta+\beta\varepsilon-2)-\beta(1-c-\beta))+r\big(c^2-1+4\alpha\delta-c(2\alpha+\beta)\delta+\beta(1+\varepsilon+2\delta(1-\alpha-\beta-\alpha\varepsilon))\big)\big)+2(1-r)\big(r^3\alpha(\beta+\beta\varepsilon-2)+r\big(6\alpha^2\delta+2\beta(c+\beta+b\delta)+\alpha(2c+\beta(3+\delta+3\varepsilon)-6)\big)-4\alpha^2\delta-2b(1-c)\beta\delta-r^2\big(2\beta^2+2\alpha^2\delta+\alpha(\beta(4+\delta+4\varepsilon)-8)\big)\big)\tau+(1-r)\big(2r(b(c-(3-r)(1-r))+\alpha-r\alpha)\beta-(4-r)(2-r)(1-r)r\alpha^2-4b(2-r)(1-r)\alpha\beta\delta\big)\tau^2-b(4-r)(2-r)(1-r)^2r\alpha\beta\tau^3\big)$.

### A.3 Proof of Proposition 1

*Proof.* We can derive the first-order derivative of $p_F^*$, $q_F^*$, $p_P^*$, and $q_P^*$ with respect to $c$ as follows.

$\frac{\partial p_F^*}{\partial c} = \frac{2c(1-r)+(1-r)^2(\beta+\beta\varepsilon-1)-c^2}{(1-r)(1-c-r)^2\beta}$. Since $0 < N_{aF}^* = \frac{1-c-r}{(1-r)\beta} < 1$, $\frac{\partial p_F^*}{\partial c} > 0$.

$$\frac{\partial q_F^*}{\partial c} = \frac{(1-r)(1+\varepsilon)}{(1-c-r)^2} > 0.$$

$\frac{\partial p_P^*}{\partial c} = \frac{\Gamma(c)}{2(1-r)\beta((1-r)(2+(2-r)\alpha\tau)-2c)^2}$, where $\Gamma(c) = -8c^2+8c(1-r)(2+(2-r)\alpha\tau)-2(1-r)^2\big((2+(2-r)\alpha\tau)^2-\beta(2+(1-r)\alpha\tau)(2+2\varepsilon+\tau(2-b(2-r)\tau))\big)$. Because $0 < r < 1$, the quadratic coefficient of $\Gamma(c)$ is $<0$ while the monomial coefficient of $\Gamma(c)$ is greater than 0. The constraint $0 < N_{aP}^* = \frac{2(1-r)-2c+(1-r)(2-r)\alpha\tau}{2\beta(1-r)} < 1$ derives $\frac{1}{2}(1-r)(2-2\beta+(2-r)\alpha\tau) < c < \frac{1}{2}(1-r)(2+(2-r)\alpha\tau)$. Since $\Gamma(c)$ is greater than 0 when $c$ equals either $\frac{1}{2}(1-r)(2-2\beta+(2-r)\alpha\tau)$ or $\frac{1}{2}(1-r)(2+(2-r)\alpha\tau)$, therefore $p_P^*$ increases as $c$ increases.

$\frac{\partial q_P^*}{\partial c} = \frac{2(1-r)(2+2\varepsilon+\tau(2-b(2-r)\tau))}{((1-r)(2+(2-r)\alpha\tau)-2c)^2}$. Because $0 < N_{aF}^* < 1$ and $0 < N_{v2P}^* = \alpha q_P^* - b\tau \le 1$, $\frac{\partial q_P^*}{\partial c} > 0$.

### A.4 Proof of Proposition 2

Proof. $p_P^* - p_F^* = \frac{\tau * F_1(c)}{2(1-c-r)\beta((1-r)(2+(2-r)\alpha\tau)-2c)}$, where $F_1(c) = A_1c^2+B_1c+C_1$, $A_1 = 2(r\alpha+b\beta\tau)$, $B_1 = (1-r)\big(r^2\alpha\tau(\alpha+b\beta\tau)-2\beta(2+\alpha(1+\varepsilon+\tau))-2r\big(2\alpha+\alpha^2\tau+b\beta\tau-\alpha\beta\big(1+\varepsilon+\tau-b\tau^2\big)\big)\big)$, and $C_1 = -(1-r)^2\big(-4\beta+2\alpha(-r+\beta+\beta\varepsilon)+(-2+r)r\alpha^2\tau+2(-1+r)(-b+\alpha)\beta\tau+b(-2+r)r\alpha\beta\tau^2\big)$. Because $0 < N_{aF}^* < 1$, the denominator of $p_P^* - p_F^*$ is greater than 0. Besides, since $0 < N_{v2F}^* = \alpha q_F^* \le 1$ and $0 < N_{v2P}^* \le 1$, both $A_1$ and $C_1$ of $F_1(c)$ are greater than 0 and $B_1$ is $<0$. The constraint $0 < N_{aF}^* < 1$ derives $(1-r)(1-\beta) < c < 1-r$. When $c = 1-r$, $F_1(c) = -2(2-r)(1-r)^2\alpha\beta(1+\varepsilon) < 0$. This indicates that the root in the feasible region for $F_1(c) = 0$ is $c_1 = \frac{-B_1-\sqrt{B_1^2-4A_1C_1}}{2A_1}$. Therefore, when $c < c_1$, $p_P^* > p_F^*$; otherwise, $p_P^* \le p_F^*$.

### A.5 Proof of Proposition 3

Proof. $q_P^* - q_F^* = \frac{(1-r)\tau * \Theta_1(\tau)}{(1-c-r)((1-r)(2+(2-r)\alpha\tau)-2c)}$, where $\Theta_1(\tau) = -b(2-r)(1-c-r)\tau - 2c + (1-r)(2-(2-r)\alpha(1+\varepsilon))$. Because $0 < N_{aF}^* < 1$, the monomial coefficient of $\Theta_1(\tau)$ is <0. Solving $\Theta_1(\tau) = 0$ yields $\tau_1 = \frac{(1-r)(2-(2-r)\alpha(1+\varepsilon))-2c}{b(2-r)(1-c-r)}$. Therefore, if $\tau < \tau_1$ then $q_P^* > q_F^*$; if $\tau \geq \tau_1$ then $q_P^* \leq q_F^*$.

### A.6 Proof of Proposition 4

Proof. $\Pi_P^* - \Pi_F^* = \frac{\tau * F_2(c)}{4(1-r)\beta^2}$, where $F_2(c) = (4r(\alpha+\beta) + 2b\beta(2\delta + r\tau))c - (1-r)\big(r^3\alpha\tau(\alpha + b\beta\tau) + 4\delta(b\beta + 2\alpha(\alpha + b\beta\tau)) - 2r^2\big(2\alpha + 3\alpha^2\tau + b\beta\tau - \alpha\beta\big(1 + \varepsilon - 3b\tau^2\big)\big) - 2r\big(2\alpha^2(\delta - 2\tau) + \beta(2\beta - 3b\tau) + \alpha\big(-6 + \beta\big(3 + \delta + 3\varepsilon + \tau + 2b\delta\tau - 4b\tau^2\big)\big)\big)\big)$. Solving $F_2(c) = 0$ yields $c_2 = \frac{1}{4r(\alpha+\beta)+2b\beta(2\delta+r\tau)}\big((1-r)\big(r^3\alpha\tau(\alpha+b\beta\tau) + 4\delta(b\beta + 2\alpha(\alpha+b\beta\tau)) - 2r^2\big(2\alpha + 3\alpha^2\tau + b\beta\tau - \alpha\beta\big(1+\varepsilon - 3b\tau^2\big)\big) - 2r\big(2\alpha^2(\delta - 2\tau) + \beta(2\beta - 3b\tau) + \alpha\big(-6 + \beta\big(3 + \delta + 3\varepsilon + \tau + 2b\delta\tau - 4b\tau^2\big)\big)\big)\big)\big)$. When $c < c_2$, $\Pi_P^* < \Pi_F^*$; otherwise ($c \geq c_2$), $\Pi_P^* \geq \Pi_F^*$.

### A.7 Proof of Proposition 5

*Proof.*

(a) $N_{aP}^* - N_{aF}^* = \frac{(2-r)\alpha\tau}{2\beta} > 0$.

(b) $\pi_P^* - \pi_F^* = \frac{\tau * G(b)}{4(1-r)\beta^2}$, where $G(b) = (2-r)(1-r)\beta\tau(2c - (1-r)(2 + (2-r)\alpha\tau))b + 4(1-r)(1-c-r)\beta - \big(2 - 3r + r^2\big)^2\alpha^2\tau + 2(2-r)(1-r)\alpha(2c - (1-r)(2 - \beta(1+\varepsilon+\tau)))$. Since $0 < N_{aF}^* < 1$ and $0 < N_{v2P}^* \leq 1$, the monomial coefficient of $G(b)$ is <0 and the constant term of $G(b)$ is greater than 0. The constraint $N_{v2P}^* > 0$ derives $0 < b < \frac{\alpha(2c+(1-r)(\beta(1+\varepsilon+\tau)-2-(2-r)\alpha\tau))}{\beta\tau((1-r)(1+(2-r)\alpha\tau)-c)}$. When $b = \frac{\alpha(2c+(1-r)(\beta(1+\varepsilon+\tau)-2-(2-r)\alpha\tau))}{\beta\tau((1-r)(1+(2-r)\alpha\tau)-c)}$, $G(b) > 0$. Therefore, within the feasible region, it is always true that $\pi_P^* > \pi_F^*$.

(c) Based on Eq. (1) and the equilibrium solutions $p_F^*$, $q_F^*$, and $N_{aF}^*$, the surplus that active askers derive under the free viewership model is $S_F = \int_{p-q-\varepsilon+q\beta N_a}^{1} (q - p + \varepsilon)dc_v = \frac{(1-c-r)(c-(1-r)(1-\beta\varepsilon))}{(1-r)^2\beta^2}$. Likewise, we can derive the total utilities of active askers under the paid viewership model as $S_P = \frac{1}{4(1-r)^2\beta^2}\big((1-r)(2 + (2-r)\alpha\tau) - 2c\big)\big(2c + (1-r)\big(2\beta\varepsilon - (2-r)\alpha\tau - b(2-r)\beta\tau^2 - 2\big)\big)$. $S_P - S_F = \frac{(2-r)\tau * \Theta_2(\tau)}{4(1-r)\beta^2}$, where $\Theta_2(\tau) = A_2\tau^2 + B_2\tau + C_2$, $A_2 = -b(1-r)(2-r)\alpha\beta$, $B_2 = -(2-r)(1-r)\alpha^2 - 2b(1-c-r)\beta$, and $C_2 = 4c\alpha + 2(1-r)\alpha(\beta\varepsilon - 2)$. Because $0 < r < 1$ and $0 < N_{aF}^* < 1$, both $A_2$ and $B_2$ are <0. Solving $\Theta_2(\tau) = 0$ yields $\tau_2 = \frac{-B_2 - \sqrt{B_2^2 - 4A_2C_2}}{2A_2}$, which is the only root in the feasible region. Therefore, when $\tau < \tau_2$, $S_P > S_F$; otherwise ($\tau \geq \tau_2$), $S_P \leq S_F$.

### A.8 Proof of extension

*Proof.* When considering the substitutability between different questions and answers, the overall number of viewers attracted by these questions becomes $N_a(1 - \lambda N_a)N_{v2}$. In the free viewership model, both the profit and the optimal decisions (i.e., $\widehat{p}_F^*$ and $\widehat{q}_F^*$) of the answerer remain unchanged, so do the number of active askers (i.e., $\widehat{N}_{aF}^*$) and the total surplus of active askers (i.e., $\widehat{S}_F^*$), while the platform's profit changes to $\widehat{\Pi}_F^* = \big(\beta^2(c + \beta - 1)\delta + r^3\beta(\beta + \beta\varepsilon - 1) + r\beta\big(c^2 - 1 + 4\alpha\delta - c(2\alpha + \beta)\delta + \beta(1 + \varepsilon - 2\delta(\alpha + \beta + \alpha\varepsilon - 1))\big) - r\alpha\delta(4 - c(4-\beta) - 2\beta - (2-c)\beta\varepsilon)\lambda + \alpha\delta(2c + \beta + \beta\varepsilon - 2)(\beta - (1-c)\lambda) + r^2(\beta(2 - 2\alpha\delta - 2\beta(1+\varepsilon) + \beta\delta(\alpha + \beta + \alpha\varepsilon - 1)) - \alpha\delta(\beta + \beta\varepsilon - 2)\lambda)\big)/\big((1-r)^2\beta^3\big)$.

In the paid viewership model, the profit of the answerer changes to

$\max_{p,\,q} \widehat{\pi}_P = N_a(p(1-r) - cq) + N_a(1 - \lambda N_a)N_{v2}\,\tau\frac{1-r}{2}$, s.t. $p, q > 0$, $0 < N_a < \frac{1}{\lambda}$, $0 < N_{v2} \leq 1$.

Based on the first-order derivative of $\widehat{\pi}_P$ with respect to $p$ and $q$, we have $\widehat{p}_P^* = \big(\alpha\lambda\tau\big(4c^2 - c(1-r)\big(16 - 2\alpha(4r + \lambda + \varepsilon\lambda - 6)\tau - 2(2b + \alpha)\lambda\tau^2 - b(1-r)\alpha\lambda\tau^3\big) + (1-r)^2\big(12 + \tau\big(\alpha\big((2-r)(4-3r)\alpha + ((2r-3)\alpha(1+\varepsilon) - 4)\lambda\big)\tau - 4\alpha(3r + \lambda - \varepsilon\lambda - 5) - 4b\lambda\tau + (2r-3)\alpha(b+\alpha)\lambda\tau^2\big)\big) - 2(1-r)\beta^2\big(4(1-r)(1+\varepsilon) + 2(1-r)(2 + (1-r)\alpha(1+\varepsilon))\tau - 2\big(b\big(c + (1-r)^2\big) - (1-r)^2\alpha\big)\tau^2 + b(2-r)(1-r)r\alpha\tau^3\big) - \beta\big(8c^2\big(b\lambda\tau^2 - 1\big) + 2c(1-r)\tau(\alpha(4 + \lambda(2 + 2\varepsilon + \tau(2 - 3b(2-r)\tau))) - 4b\lambda\tau) + (1-r)^2\big(8 + \alpha\tau\big(8 - 4\lambda(3 + 2\alpha\tau)\big(1 + \varepsilon + \tau - b\tau^2\big)\big) + r^2\alpha\tau\big(2 + 3b\lambda\tau^2\big) + 2r\big(\alpha\tau(\lambda(3 - 3\varepsilon + \tau(3 - 5b\tau)) - 2) - 4 - 3b\lambda\tau^2\big)\big)\big)\big)\big)/\big((1-r)(2\beta + \alpha\lambda\tau)^2(2c - (1-r)(2 + (2-r)\alpha\tau))\big)$ and $\widehat{q}_P^* = \frac{4c\big(b\lambda\tau^2 - 2\big) + (1-r)(8 + 4(2-r)\alpha\tau - 2\beta(2 + 2\varepsilon + \tau(2 - b(2-r)\tau)) - \lambda\tau(4b\tau + \alpha(2 + 2\varepsilon + \tau(2 + b(2-r)\tau))))}{(2\beta + \alpha\lambda\tau)(2c - (1-r)(2 + (2-r)\alpha\tau))}$. Then, the optimal profits of the answerer and the platform can be derived as $\widehat{\pi}_P^* = \big((2c - (1-r)(2 + (2-r)\alpha\tau))\big(c\big(b\lambda\tau^2 - 2\big) - (1-r)\big(\alpha(r + \lambda + \varepsilon\lambda - 2)\tau + (b + \alpha)\lambda\tau^2 + \beta(2 + 2\varepsilon + \tau(2 - b(2-r)\tau)) - 2\big)\big)\big)/\big((1-r)(2\beta + \alpha\lambda\tau)^2\big)$ and $\widehat{\Pi}_P^* = \widehat{N}_{aP}^*\widehat{p}_P^* r + \big(\widehat{N}_{aP}^*\big(1 - \lambda\widehat{N}_{aP}^*\big)\big(\alpha\widehat{q}_P^* - b\tau\big) + \big(1 - \widehat{N}_{aP}^*\big)\big)(\delta + \tau r)$, respectively. The number of active askers in the paid viewership model becomes $\widehat{N}_{aP}^* = \frac{(1-r)(2 + (2-r)\alpha\tau) - 2c}{(1-r)(2\beta + \alpha\lambda\tau)}$, and their total surplus is $\widehat{S}_P^* = \big(((1-r)(2 + (2-r)\alpha\tau) - 2c)\big(2(1-r)\beta^2\big(2\varepsilon - b(2-r)\tau^2\big) + \beta\big(c\big(4 - 4b\lambda\tau^2\big) - (1-r)\big(4 + 2\alpha(2 - r + (1-\varepsilon)\lambda)\tau + 2(\alpha - 2b)\lambda\tau^2 - b(2-r)\alpha\lambda\tau^3\big)\big) + \alpha\lambda\tau((1-r)(2 - \alpha\tau(r + \lambda + \lambda\tau - 2)) - 2c)\big)\big)/\big((1-r)^2(2\beta + \alpha\lambda\tau)^3\big)$.

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**Haiyang Feng** is a professor of Information Management and Management Science at the College of Management and Economics, Tianjin University. He received his Ph.D. in Management Science from Tianjin University in 2014. His current research interests include economics of information systems, platform strategy, and business analytics. His papers have been published in academic journals including *MIS Quarterly, Journal of the Association for Information Systems, Decision Support Systems, Annals of Tourism Research, International Journal of Production Economics*, among others.

**Muzi Li** studied at the College of Management and Economics, Tianjin University, and received a master's degree in Management Science in 2023. Her research direction is information system economics.

**Wenfan Zhao** is a lecturer of Human Resource Management and Public Administration at the Faculty of Humanities and Social Sciences, Beijing Institute of Petrochemical Technology. She received her Ph.D. in Management Science from Tianjin University in 2023. Her research interests include economics of information systems, social media and user-generated content. Her recent papers have appeared in *Journal of Management Science and Engineering, International Journal of Computer Integrated Manufacturing, Kybernetes*, among others.

**Nan Feng** is a professor of Information Management and Management Science at the College of Management and Economics, Tianjin University. He received his Ph.D. degree in Management Science from Tianjin University in 2007. His current research interests include economics of information systems, information security, and business analytics. *He has published in MIS Quarterly, Decision Support Systems, Information & Management, Electronic Commerce Research and Applications, Enterprise Information Systems*, among others.